

Introduction to Differential Equations

Date _____ Period _____

Find the general solution of each differential equation.

1) $\frac{dy}{dx} = 2x + 2$

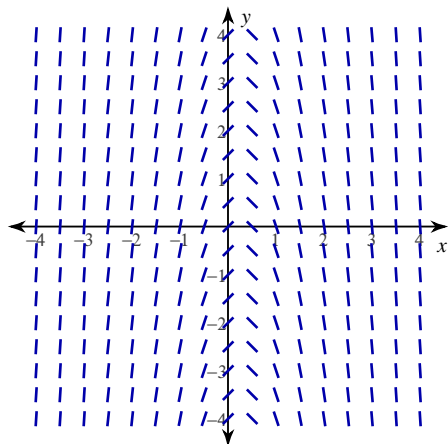
2) $f'(x) = -2x + 1$

3) $\frac{dy}{dx} = -\frac{1}{x^2}$

4) $\frac{dy}{dx} = \frac{1}{(x+3)^2}$

For each problem, find the particular solution of the differential equation that satisfies the initial condition. You may use a graphing calculator to sketch the solution on the provided graph.

5) $\frac{dy}{dx} = -4x + 1, y(-1) = -3$



6) $\frac{dy}{dx} = -2\sin x, y\left(\frac{2\pi}{3}\right) = 0$

